Chapter 1: Sectoral Convergence in the United States: 1800-2020

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February 26, 2026

Abstract

DISCLAIMER: This is a working paper that will be revised through my dissertation defense. The most current version is available at:
`https://sites.google.com/view/jacob-campbell-economics/research-career?authuser=2`

This paper investigates the dynamics of productivity convergence across the United States from 1800 to 2020. Exploiting a novel historical dataset from Tamura, Witham and Khatri (2025, working paper) I examine β- and σ-convergence at both aggregate and sectoral levels for the entire history of the United States. I find a long-run trend toward convergence consistent with the conditions of an integrated national market with free mobility of labor and capital. However, the pace of convergence varies considerably, being weak before 1860, strengthening by 1910 and robust thereafter. Sectoral results demonstrate strong and persistent convergence in manufacturing throughout the period, while agricultural and service sector convergence patterns differ markedly, with services showing the strongest convergence post-1910. These patterns suggest that different sectors drove aggregate convergence at different times, with manufacturing convergence dominant in the 19th century and service sector convergence dominant in the 20th century.

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1 Introduction and Motivation

The income covergence hypothesis - the question of whether poor regions catch up to rich regions - is one of the fundamental results of growth theory. Beginning with Solow (1956) and Swan (1956), diminishing returns to physical capital combined with depreciation of the capital stock imply that smaller economies will have disproportionately higher marginal rates of investment and so ultimately grow to match their richer neighbors. However, the empirical reality has proven more complex, with convergence often being demonstrated to be conditional or convergence occuring within ”clubs” of similar nations.

To this extent the United States status as the largest industrialized federal republic makes it an ideal laboratory for natural experiments in regional convergence. Sharing a common language, legal system, currency and federal government American states fewer institutional barriers to factor mobility while also sharing the same sorts of institutions that Acemoglu recognized as being a primary determinant of economic growth. Barro and Sala-i-Martin (1992) became the canonical benchmark for U.S. convergence patterns, finding robust convergence over the period 1880-1990 with an estimated semi-elasticity (growth rate percentage points with respect to initial real income) of 2 percentage points per year.

However, two gaps limit our understanding of U.S. regional convergence. One is that existing studies have largely focused - for reasons of lack of quality historical data - on the 1880s onward, leaving the earlier dynamics of the nineteenth century neglected. For a convergence paper this is problematic, as the initial divergence in incomes would be expected to have occurred over the period of the First Industrial Revolution, the Commercial Revolution, slavery and abolition, westward expansion and urbanization as these historical processes operated differently from region to region. In short, if the existing literature mostly explores convergence of income among the states, the initial divergence - or potentially lack thereof - is relatively unexplored. Was convergence present from the beginning, or did it accompany the development of integrated national markets and the expansion of transportation services? Likewise, aggregate convergence patterns can mask very different causal mechanisms at the sectoral level. The existing structural transformation literature recognizes that aggregate convergence and divergence may be driven by sectoral reallocation of labor - such as industrialization and later tertialization - as much as by real differences in productivity.

This paper addresses both gaps by examining productivity convergence across the United States from 1800-2020, disaggregated by three sectors (agricultural, industrial and services). I exploit a novel dataset introduced in Tamura, Witham and Khatri (2025) that reconstructs state-level output per worker for each of these sectors extending back to 1800, a full 80 years earlier than Barro and Sala-i-Martin (1992)'s seminal paper. This extended panel allows me to capture the full course of American economic development from a pre-War of 1812 agrarian economy, through the beginnings of the Industrial Revolution and the formation a national market as regional patterns would have been expected to develop. I employ both standard tests of β-convergence (where initial income negatively predicts subsequent growth) and σ-convergence (where cross-sectional dispersion narrows over time) at both aggregate and sectoral levels. Following the methodological critiques of Quah (1993, 1996a, 1996b), Bernard and Durlauf (1996), and Evans and Karras (1996), I recognize that β-convergence alone provides insufficient evidence for true economic catch-up. I therefore emphasize σ-convergence as the more reliable indicator while using β-estimates to quantify convergence speeds and investigate temporal variation.

My findings reveal robust patterns of weak early β-convergence and strong early σ-divergence across sectors and states. At the aggregate level, I find evidence of long-run convergence consistent with an integrated national market, but the pace and pattern shifted markedly over time. Convergence was weak and uncertain before 1860, strengthened considerably by 1910, and became robust thereafter. Cross-sectional dispersion in aggregate output per worker peaked around 1850-1890 before declining steadily through most of the twentieth century, with a brief reversal around 1970.

The sectoral results demonstrate that this aggregate pattern masked substantial heterogeneity. Manufacturing productivity exhibited strong and persistent convergence throughout the entire 1800-2020 period, with β-coefficients consistently in the 2-3% range—suggesting that industrial technology and capital deepening diffused relatively uniformly across regions even in the early nineteenth century. Agricultural convergence, by contrast, was weak in the antebellum period, strengthened from 1850-1960, but then reversed dramatically after 1960. The service sector showed the most dramatic evolution: negligible convergence before 1850, followed by explosive convergence from 1850-1990, and then modest divergence in recent decades.These sectoral patterns suggest that different sectors drove aggregate convergence at different times. In the nineteenth century, manufacturing convergence dominated aggregate trends. In the twentieth century, service sector convergence became the primary driver. The recent slowdown in aggregate convergence appears linked to the exhaustion of service sector catch-up and the reversal of agricultural convergence.

My findings contribute to several literatures. First, they extend the U.S. convergence literature backward into the early nineteenth century, revealing that convergence mechanisms operated differently - and more weakly - before the industrialization of the country. Second, the sectoral results connect to the structural transformation literature (Herrendorf, Rogerson and Valentinyi 2014) by showing that aggregate convergence reflects both within-sector productivity convergence and labor reallocation across sectors. The interaction between these forces shaped regional income dynamics in ways not captured by aggregate analysis alone. Third, my findings parallel Polovnikov's (2023) results for OECD countries, which showed aggregate divergence before 1910 followed by convergence, despite sectoral convergence throughout. While I do not find the same stark aggregate divergence in the more institutionally homogeneous U.S. context, I do find that sectoral convergence patterns varied considerably across time, with resulting effects on aggregate trends.

This paper continues in five sections. Section 2 reviews the theoretical and empirical literature on convergence, with particular attention to methodological debates and sectoral heterogeneity. Section 3 describes the novel dataset and its construction. Section 4 presents the main empirical results, first at the aggregate level and then disaggregated by sector. Section 5 discusses future research directions, and Section 6 concludes.

2 Literature Review

The concept of economic convergence - whether poor regions catch up to rich regions in time - has been one of the central questions of growth economics since the development of the Solow-Swan model. In Solow-Swan, convergence is a consequence of diminishing returns to capital. Conditional on a common technological level, poorer economies should experience higher marginal savings, leading to their capital deepening at a faster rate than more advanced economies and the output gap closing. Barro and Sala-i-Martin (1992) provided the standard empirical test and estimates. Termed "β-convergence," the argument goes that initial income should negatively correlate with subsequent growth rates. Their work on both US states and European regions predicted a semi-elasticity of 0.01 to 0.03: that is, that a 1% decrease in the initial level of income would correspond to a 0.01 to 0.03 percentage point increase in the growth rate of real output per worker. This estimate became the benchmark for the subsequent convergence literature.

However, an extensive methodological literature formed demonstrating that β-convergence is insufficient evidence for convergence in the conventional, Solow-theoretic sense. Quah (1993) introduced the concept of "Galton's fallacy," arguing that a negative β coefficient may reflect regression to the mean rather than catching up. His later work (1996a, 1996b) emphasized that cross-sectional regressions hide distributional dynamics, particularly with respect to the possibility of convergence clubs. Bernard and Durlauf (1996) and Sala-i-Martin (1996) demonstrated that a negative β coefficient is neither necessary nor sufficient for convergence, and Evans and Karras (1996) demonstrated that heterogeneous steady states or adjustment speeds render cross-sectional β estimates fragile.

These critiques motivated the development of an alternate concept of σ-convergence: convergence defined as the narrowing of cross-sectional income dispersion. Wodon and Yitzhaki (2006) proved algebraically that σ-convergence is sufficient for β-convergence, rendering the point of β-convergence moot except for the utility of measuring the rate of convergence. This paper follows this dual approach, presenting σ-convergence evidence (for various balanced panels and one unbalanced panel) along with both cross-sectional and panel estimates of β. Notably, Wodon and Yitzhaki's result only holds for balanced panels; otherwise, the addition of new regions to the panel can drive σ-divergence even as β-convergence holds.

The empirical literature on US state convergence has, as in Barro and Sala-i-Martin (1992), largely focused on the period from 1880 onward with earliest estimates from 1840 with a special focus on the "Great Compression" of the 1940s-1970s when regional wage and income gaps narrowed dramatically. However, the early period of the United States has been relatively neglected due to census data quality issues. This paper seeks to contribute to the existing literature by exploiting a novel dataset introduced in Tamura, Witham and Khatri (2025) that reconstructs estimates of output per worker and extends coverage back to 1800, giving rare empirical insight into the American First Industrial Revolution and Market Revolution when distinctive regional patterns of development first emerged.

Beyond temporal coverage, a critical gap in the convergence literature concerns sectoral heterogeneity. Caselli and Coleman (2001) demonstrated that structural transformation—the reallocation of labor from agriculture to manufacturing and services—plays a central role in regional convergence. They showed that much of the apparent productivity convergence across US regions reflected the movement of workers out of low-productivity agriculture rather than within-sector productivity gains. This insight connects convergence research to the broader literature on structural transformation (Herrendorf, Rogerson and Valentinyi 2014), which emphasizes that aggregate productivity growth depends not only on sector-specific technical progress but also on the reallocation of resources across sectors with different productivity levels. More recently, Polovnikov (2023) documented that OECD countries experienced a pattern of aggregate divergence before 1910 followed by aggregate convergence afterwards despite convergence at the sectoral level. Despite the existence of the structural transformation literature, lack of data has tended to limit sectoral disaggregation to more recent decades. Another contribution of the new dataset introduced in

Tamura, Witham and Khatri (2025) and used in this paper is that it disaggregates to the sectoral level (agriculture, industry and services) for the entire 1800-2020 period and for a much more homogeneous set of regions (US states) that share common institutions, cultural origins and a high degree of both capital and labor mobility. While I do not find such a clear pattern of divergence and then convergence, I find evidence of sectoral convergence strengthening at different times, with resulting effects on aggregate convergence.

3 Data

For my empirical analysis, I draw on novel decadal panel data introduced in Tamura, Witham and Khatri (2025), which track output per worker and key factors of production over the period 1800-2020 at the US state level. The dataset is of interest for two reasons. First is that it extends coverage to the early nineteenth century (1800-1840), a period rarely included in growth-economic studies due to poor quality of available data. As a new source of high quality data, this allows for earlier transition dynamics to be included in analysis. Second, it disaggregates the data by sector - agriculture, industry and services - across the entire sample. Together, the earlier period and the sectoral decomposition are valuable as they reveal a marked shift in convergence patterns prior to 1850-1860. Because the number of states is small at the start of the sample (growing from 16 to 50), I present results for both unbalanced panels and panels balanced on alternative start years, alternative periodizations and cross-sectional results. Details on data sources, construction, and summary statistics (including the earliest decade of coverage for each state) can be found in Appendix A.

4 Empirical Analysis

In this section, I present the empirical analysis of the convergence hypothesis among the United States. The two most common convergence tests are the β-convergence test and the σ-convergence test. β-convergence, the simpler of the two, is an implication of the neoclassical growth model introduced in Solow-Swan (1956). If countries share the same level of technology and savings rate, then the diminishing marginal product of capital combined with depreciation of said capital will lead developing economies (with a higher net marginal output) to accumulate capital faster and, hence, converge to developed economies.

Stated in log-linearized form, the standard statistical test is:

$$\frac{log(y_{iT}/y_{i0})}{T} = \alpha + \beta log(y_{i0}) + \epsilon_i \tag{1}$$

in which y_{iT} is real output per worker in state i in period T and y_{i0} is real output per worker in the starting period in state i. If the null hypothesis that $\beta = 0$ is rejected and the alternative hypothesis is that $\beta < 0$, then the formula is consistent with the prediction that higher initial income levels reduce the growth rate of output per worker, or equivalently, lower initial income levels raise it.

However, an extensive literature has formed proving that β-convergence is not sufficient to prove the convergence hypothesis. Barro and Salah-i-Martin (1992) recognized that β-convergence is insufficient. Quah (1993, 1996a, 1996b) emphasized that a negative β coefficient may be an instance of regression to the mean rather than a narrowing of income differences. Bernard and Durlauf (1996) formalized this point in a time-series framework, demonstrating that $\beta < 0$ is neither necessary nor sufficient for output levels to converge; developing countries may grow faster and yet still fail to grow fast enough. Evans and Karras (1996) likewise found that allowing for heterogeneous steady states and adjustment speeds renders cross-sectional β estimates fragile.

If β-convergence is insufficient by itself, σ-convergence in combination with it provides evidence. σ-convergence merely focuses on the dispersion of the variable in question, arguing that if:

$$\sigma_T > \sigma_{T+S} \tag{2}$$

for every period T and length of time S, then the sample is becoming more alike. Wodon and Yitzhaki (2006) proved algebraically that σ-convergence is sufficient, as it holding implies that β-convergence will also hold. Wodon and Yitzhaki's result is still open to the counterargument that convergence could take the form of developed nations becoming undeveloped rather than the developed nations "catching up"; however, for this sample of US states, Kaldor fact 1 - that output per worker rises in the long term - holds. As such, I present σ-convergence and β-convergence results first at the aggregate level and then at the sectoral levels.

4.1 Aggregate convergence

As the earlier and more standard test, I present β-convergence first. Table 1 reports the corresponding β-convergence regressions for both panel and cross-sectional data for the periods 1800-1910 (before and during the Industrial Revolution) and 1910-2020 (after the Industrial Revolution). For the panel data the specification is:

$$\text{Specification 1: } \frac{log(y_{iT}/y_{iT-10})}{10} = \alpha + \beta log(y_{iT-10}) + \epsilon_i \tag{3}$$

While the cross-sectional (long-run) regressions are:

$$\text{Specification 2: } \frac{log(y_{ij_2}/y_{ij_1})}{j_2 - j_1} = \alpha + \beta log(y_{ij_1}) + \epsilon_i \tag{4}$$

for $j \in \{(1800, 1910), (1910, 2020)\}$.

The resuts of Table 1 show weak evidence of β-convergence for specification (1) for both unbalanced and balanced panels and for specification (2) for each period. Across specification (1), both periods report highly statistically significant and economically significant convergence trends robust to the use of balanced panel data; for specification (2), the result only holds for the later period. The β estimates are consistent with the range commonly reported in the US convergence literature of 1% to 3% established in Barro and Salah-i-Martin (1992). The exception is the balanced panel for 1800-1910, which suffers from a very small and less geographically representative sample size. Evidence also exists, however, for an increase in the rate of convergence in the later period, with the magnitude doubling-to-quadrupling. This result supports the findings of Polovnikov (2023), who found that among OECD countries over a period of 1840-2020 there was first a pattern of aggregate divergence before 1910, and then aggregate convergence afterwards. Among a cluster of economies with more similar institutions - shared forms of government, integrated markets, a common language, a common legal tradition - it is both predictable that the divergence pattern would not hold, but also notable that convergence would intensify over time.

	1800-1910			1910-2020		
	Unbal. panel	Bal. panel	Cross-section	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (aggregate) (decadal)	-0.0145***	-0.0057***		-0.0225***	-0.0222***	
	(0.0031)	(0.0019)		(0.0023)	(0.0025)	
Initial Output per worker (aggregate) (long-run)			0.0028			-0.0080***
			(0.0039)			(0.0005)
Observations	350	176	16	488	470	47
R^2	0.47	0.53	0.04	0.43	0.46	0.87

Standard errors in parentheses
Notes:
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 1: β-convergence, 1800-1910 and 1910-2020

Considering visual evidence, Figure 1 plots average annual growth rates in aggregate output per worker against the logged initial output per worker for both periods at a decadal level. Both periods demonstrate a negative correlation, but with a considerable deal of spread and heteroskedasticity in the 1910-2020 period among low-income states. Moreover, the correlation is stronger in 1800-1910, in contradiction to the lower observed estimates of β.

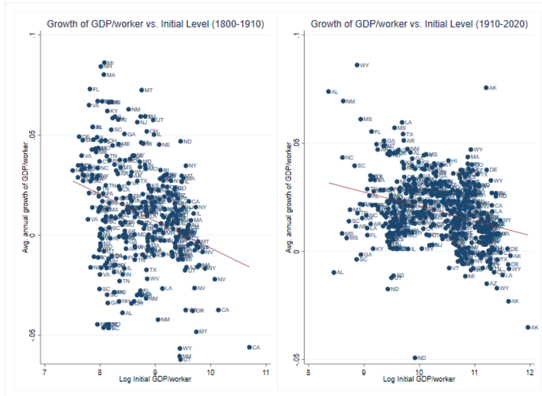

Figure 1: β-convergence scatterplot, decadal intervals, 1800-1910 and 1910-2020

For cross-sectional data (average annual growth rates over 110 years each), growth rates and initial output per worker are strongly negatively correlated past 1910, but positively correlated before 1910, consistent with the results reported in Table 1 for specification (2).

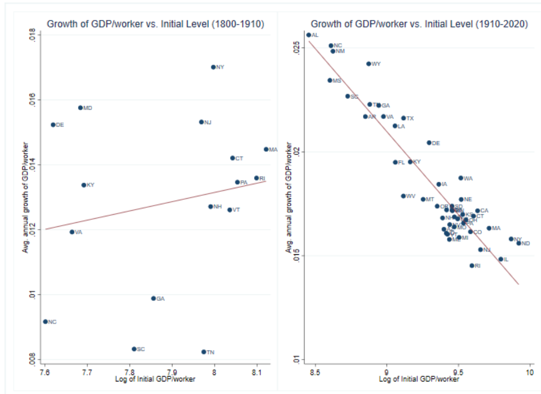

Figure 2: β-convergence scatterplot, long-run, 1800-1910 and 1910-2020

As noted, β-convergence is insufficient to prove convergence in the traditional sense. Figure 3 presents results for σ-convergence from 1800-2020. Due to the rapidly changing composition of US states over the time period, one may be concerned about the standard deviation being a noisy variable driven more by the addition of new geographical clusters of states, with their own economic characteristics, rather than change among the existing states. As a robustness I present four panels: an unbalanced panel and panels balanced to years 1800, 1850 and 1880, containing 16, 32 and 47 states respectively, or roughly one-third, two-thirds, and almost all states represented in the panel. A table of coverage can be found in Appendix A.1 Figure 6. From 1950 on, all 50 states are covered.

Figure 3 demonstrates, across all panels, heavy σ-divergence up to 1850, stagnation to 1890 and gradual, non-monotonic convergence to the present with a brief reverse around 1970. For the period 1910-2020 the σ-convergence trends support our β-convergence results of steady convergence among the states; before 1910, however, it is at direct odds with the results of Table 1, suggesting weak evidence. In the Robustness section I present alternate periodizations for a separate 1850-1910 period and a split panel around 1890.

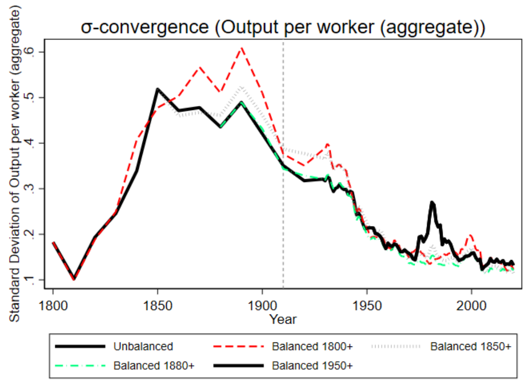

Figure 3: σ-convergence of output per worker for each sector

Combining the interpretation of Figure 3 with Table 1, there is conflicting evidence of divergence prior to 1860 and strong evidence of convergence afterwards consistent with the existing literature.

4.2 Sectoral convergence

Aggregate results mask important differences across sectors. Because the period 1800-2020 covers the full historical spectrum of agrarianism, industrialization and tertialization, changes in aggregate output per worker and its convergence may be driven by heterogeneous paces of sectoral reallocation as well as by actual convergence in sectoral-level productivity; Figure 4 shows the evolution of labor force share allocated to each sector at the level of the census division. In this section I report convergence results at the sectoral level for agriculture, industry (manufacturing, mining and construction) and services (all other sectors).

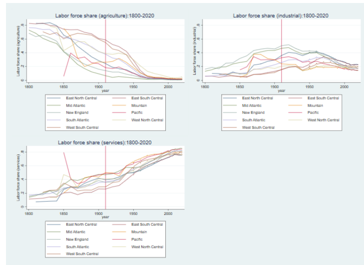

Figure 4: Labor force share in each sector, 1800-2020, Census division level

Table 2 reports the results of specifications (1) and (2) for agricultural output per worker. Across agriculture, evidence for convergence strengthens for 1800-1910's long-run cross-section and otherwise holds the same high statistical significance and economic significance. There is a reversal of trend, however, between specification (1) and specification (2) as cross-sectional data shows stronger convergence in the earlier period and panel data shows stronger convergence in the later period.

	1800-1910			1910-2020		
	Unbal. panel	Bal. panel	Cross-section	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (agriculture) (decadal)	-0.0097***	-0.0176***		-0.0212***	-0.0215***	
	(0.0023)	(0.0040)		(0.0026)	(0.0026)	
Initial Output per worker (agriculture) (long-run)			-0.0174**			-0.0077***
			(0.0067)			(0.0015)
Observations	350	176	16	488	470	47
R^2	0.46	0.49	0.33	0.56	0.56	0.38

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2: β-convergence in agriculture, 1800-1910 and 1910-2020

Results for industrial output per worker are shown in Table 3. In contrast to agriculture's conflicting pattern, industry is seen across all specifications consistently converging and at the higher end of typical estimates, suggesting strong and early convergence among the states.

	1800-1910			1910-2020		
	Unbal. panel	Bal. panel	Cross-section	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (industrial) (decadal)	-0.0304***	-0.0310***		-0.0299***	-0.0298***	
	(0.0040)	(0.0071)		(0.0045)	(0.0047)	
Initial Output per worker (industrial) (long-run)			-0.0025			-0.0113***
			(0.0022)			(0.0010)
Observations	350	176	16	488	470	47
R^2	0.55	0.57	0.08	0.65	0.66	0.74

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: β-convergence in industry, 1800-1910 and 1910-2020

Lastly, results for services output per worker are shown in Table 4. Services most match the aggregate picture, showing highly statistically significant convergence across all specifications (except for the early cross-section) and a pattern of intense early convergence followed by even higher convergence past 1910.

	1800-1910			1910-2020		
	Unbal. panel	Bal. panel	Cross-section	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (services) (decadal)	-0.0283***	-0.0153***		-0.0406***	-0.0420***	
	(0.0034)	(0.0026)		(0.0040)	(0.0041)	
Initial Output per worker (services) (long-run)			-0.0024			-0.0087***
			(0.0032)			(0.0006)
Observations	350	176	16	488	470	47
R^2	0.32	0.26	0.04	0.61	0.64	0.81

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: β-convergence in services, 1800-1910 and 1910-2020

Collectively, the results show three sectors all converging across the entire 1800-2020 panel, but with differing rates and patterns. For agriculture, β-convergence from decade to decade begins weak and strengthens only after industrialization. For industry, it remains consistent throughout the period. For services, it too strengthens like agriculture, but begins fast and speeds up even more. Consistent with our pattern of early industrialization followed by tertialization, it is expected that aggregate convergence may be driven by both convergence in sectoral productivity levels and in sectoral reallocation from laggard sectors to high-productivity sectors, reflecting increasing specialization in state-level economies.

Figure 5 reports the σ-convergence for each sector side-by-side. Unlike the dispersion of aggregate output per worker, sectoral output per worker shows a unique pattern for each sector. In agriculture, divergence grows logarithmically, peaks around 1910, and then converges to about 1960 before beginning an unstable divergence to both a global and local maximum. In industry, convergence and divergence trends experience frequent reversals with an overall but less significant trend towards convergence from roughly 1880 to the present. Lastly, the service sector experiences dramatic σ-divergence from 1830 to 1850, roughly, followed by as rapid of divergence and then a more gradual decline to the present.

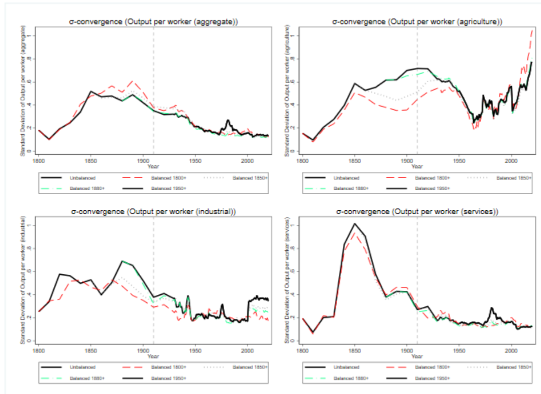

Figure 5: σ-convergence of output per worker for each sector (unbalanced panel)

As with the aggregate results, σ-convergence does not necessarily support the results of our β-convergence analysis. Across all sectors there is strong σ-divergence in the early period followed by - in agriculture and mining - stagnation and finally convergence; for agriculture, the convergence trend entirely reverses by the end of the period.

5 Future Direction

While this paper establishes the patterns of convergence across sectors over time, it does not provide a causal mechanism. Future research will need to address two main results: why there is early and simultaneous β-convergence and σ-divergence, and the cause of the sectoral heterogeneity with respect to timing and magnitude. As previously noted, Figure 5 demonstrates that even balanced panels showed convergence patterns over time that reasonably tracked the unbalanced panel. One possible cause of σ-divergence is changing composition of the panel, but that cannot be the case here. However, other mechanisms may be at play. Preliminary analysis of rank mobility (measured as mean absolute percentile change from decade to decade in state income ranks) finds high leapfrogging in the early period of 1800-1840, with states experiencing an average rank change of 20-30 percentile points between 1800 and 1840 in non-service sectors. This suggests that the observed pattern of β-convergence with σ-divergence is capturing a great deal of churn where smaller economies not only grow but overshoot rich economies, creating a pattern that is consistent with the hypothesis of small economies growing faster but inconsistent with the hypothesis that similar institutions and technology levels yield similar steady states. Robustness checks in Appendix B.1.2 provide further evidence of this hypothesis. Future iterations of this research will investigate this phenomenon further in both the early period and (particularly for agriculture) late period. Transitional probability matrcies (employed by Quah (1996a, 1996b)) may be one way to identify the transition dynamics involved in the unstable early period.

Second, the motivation of studying American states is in part the great degree of factor mobility that one would expect from a federal republic with open state borders. An economic historian may expect that, consistent with the historical literature, the expansion of railroads, steam power, and canals before 1860 and later continued expansion of railroads as well as the interstate road system after 1910 would contribute heavily to convergence among a set of economies that are all interlinked in the absence of border controls or protectionist tariffs. Consistent with existing literature on economic integration, the analysis could be extended by testing rates of convergence against factors like interstate rail and canal networks, the size of lacustrine/coastal merchant fleets and (if present) price convergence as hypothesized by the Law of One Price. Collectively, these empirical tests would give insight into whether the observed patterns of convergence are driven by market integration. Likewise, if spillovers from labor migration could be measured (such as through Census-linking introduced in Ferrie (1996)), the impact of labor mobility on convergence rates would be estimable.

In addition, this paper provides estimates of β, but it does not estimate directly the contribution of the within-industry and between-industry effects of structural transformation. Future revision of this analysis will directly measure this, as seen in Polovnikov (2023)'s similar treatment of OECD nations.

6 Conclusion

This paper investigates income convergence across the United States from 1800 to 2020 at the sectoral (agriculture, industry and services) level. It contributes to the existing convergence literature by exploiting a novel dataset (introduced in Tamura, Witham and Khatri (2025)) that both extends coverage to a much earlier period (the early nineteenth century) than is customary and also disaggregating said data to the sectoral level, providing empirical evidence for a period uncovered by previous research. The results demonstrate evidence for β-convergence over the entire period, but σ-divergence in the earliest decades that only the extended coverage shows. Additionally, the pace of convergence, as measured by estimates of β, varies over sector and time. While β-coefficient estimates largely fall within/around literature benchmarks of 1-3%, in two-period regressions the pace of convergence strengthens heavily for services in the twentieth century and less so for agriculture, with industry remaining steady. Aggregate convergence patterns mask this sectoral heterogeneity, demonstrating the role that first industrialization and then tertialization may have played in driving convergence as sectoral transformation reallocated labor from agriculture to sectors that were converging faster. The findings are robust in time and space, as alternative periodizations, jack-knifing at both the state and census divisional levels and regional-level regressions do not fundamentally change the interpretation.

The analysis leaves important questions for future research. Why did the early period have simultaneous β-convergence and σ-divergence? Preliminary results suggest a pattern of leapfrogging in which relatively poor states frequently not only caught up to but overtook relatively rich states, leading to faster growth in less developed economies even as income gaps widened. The robustness of the observed σ-convergence patterns with respect to different balanced panels disproves that changes in panel composition (from the addition of new states/westward expansion) can be the cause. Likewise, while this paper documents when and where convergence occured, it does not establish causal mechanisms. Why would some sectors converge earlier or faster than others? What role does the formation of an integrated national market play in the process of convergence among fairly homogeneous states play? Future extensions of this research will address these questions, as well as estimating the contribution of within-industry and between-industry components using shift-share analysis.

By extending the analysis of US regional convergence to 1800 and disaggregating, this paper provides new evidence for convergence at an even earlier date, as well as the significance of structural transformation in explaining regional income inequalities.

7 References

Barro, R. J., & Sala-i-Martin, X. (1992). Convergence. Journal of Political Economy, 100(2), 223–251.

Bernard, A. B., & Durlauf, S. N. (1996). Interpreting tests of the Convergence Hypothesis. Journal of Econometrics, 71(1–2), 161–173.

Caselli, F., & Coleman II, W. J. (2001). The U.S. Structural Transformation and Regional Convergence: A reinterpretation. Journal of Political Economy, 109(3), 584–616.

Evans, P., & Karras, G. (1996). Do economies converge? evidence from a panel of U.S. states. The Review of Economics and Statistics, 78(3), 384–388.

Herrendorf, B., Richard, R., & Valentinyi, Á. (2014). Growth and Structural Transformation. In Handbook of Economic Growth (Vol. 2, pp. 855–941). essay, Elsevier B.V.

Polovnikov, I. (2023). Two Essays in Human Capital (dissertation).

Quah, D. (1993). Galton's fallacy and tests of the Convergence Hypothesis. The Scandinavian Journal of Economics, 95(4), 427–443.

Quah, D. T. (1996a). Empirics for economic growth and Convergence. European Economic Review, 40(6), 1353–1375.

Quah, D. T. (1996b). Twin Peaks: Growth and convergence in models of distribution dynamics. The Economic Journal, 106(437), 1045–1055.

Solow, R. M. (1956). A contribution to the theory of economic growth. The Quarterly Journal of Economics, 70(1), 65.

Swan, T. W. (1956). Economic growth and capital accumulation. Economic Record, 32(2), 334–361.

Tamura, R., Witham, A., & Khatri, K. (2025). Unpublished manuscript.

Wodon, Q., & Yitzhaki, S. (2006). Convergence forward and backward? Economics Letters, 92(1), 47–51.

A Data

A.1 Data Summary

Table 5: Summary statistics (1800-2020)

	N	Mean	SD	Min	Max
Output per worker (aggregate)	941	29588.529	27397.817	1813.537	156677.766
Output per worker (agriculture)	941	25302.754	55683.082	734.797	1065319.000
Output per worker (industrial)	941	28799.318	31226.481	689.409	336104.313
Output per worker (services)	941	32899.779	26021.421	689.973	168051.688
total labor force	941	1265111.554	1988634.342	9500.000	18713000.000
agricultural labor force	941	126822.655	134773.359	267.542	866953.063
industrial labor force	941	318024.910	476858.168	619.098	3608722.250
service labor force	941	820317.339	1560651.446	587.871	15684194.000
Labor force share (aggregate)	941	1.000	0.000	1.000	1.000
Labor force share (agriculture)	941	0.275	0.266	0.000	0.903
Labor force share (industrial)	941	0.235	0.121	0.019	0.633
Labor force share (services)	941	0.490	0.220	0.013	0.911
Earliest decade	941	1833.071	35.849	1800.000	1950.000
Average age of workers	941	32.108	5.607	18.965	44.900
Schooling years per worker (aggregate)	941	7.428	4.549	0.006	15.015
Schooling years per worker (agriculture)	941	6.639	4.099	0.005	13.983
Schooling per worker (industrial)	941	7.259	4.337	0.006	15.290
Schooling per worker (services)	941	7.748	4.548	0.007	15.043
Experience years per worker (aggregate)	760	18.883	2.022	12.107	25.829
Experience years per worker (agriculture)	941	19.469	2.424	10.286	26.130
Experience years per worker (industrial)	941	18.849	2.300	10.354	25.752
Experience years per worker (services)	941	18.360	2.173	9.563	25.248
Human capital per worker (aggregate)	760	4.439	1.751	1.724	8.428
Human capital per worker (agriculture)	941	4.267	1.791	1.722	8.738
Human capital per worker (industrial)	941	4.509	1.938	1.722	9.169
Human capital per worker (services)	941	4.714	2.074	1.722	9.295
Total factor productivity (aggregate)	760	3.404	1.899	1.386	25.286
Total factor productivity (agriculture)	941	173.403	166.176	15.823	2488.249
Total factor productivity (industrial)	941	267.948	103.070	43.220	932.710
Total factor productivity (services)	941	226.087	78.377	34.190	677.039
Capital-output ratio (aggregate)	761	2059332.489	3695668.000	10015.795	36522612.000
Capital-output ratio (agriculture)	941	3.806	2.816	0.268	23.091
Capital-output ratio (industrial)	941	1.529	0.912	0.067	7.152
Capital-output ratio (services)	941	3.371	0.846	0.373	5.608
Physical capital per worker (aggregate)	761	9.169e+10	2.363e+11	72749120.000	2.711e+12
Physical capital per worker (agriculture)	941	96043.066	186530.551	750.147	1921466.500
Physical capital per worker (industrial)	941	58548.237	100546.222	228.257	1447031.250
Physical capital per worker (services)	938	113588.241	87442.930	1415.218	620806.375

Table 6: Summary statistics (1800-2020)

	N	Mean	SD	Min	Max
Output per worker (aggregate)	397	7997.195	4854.745	1813.537	44209.555
Output per worker (agriculture)	397	4593.970	4095.913	1014.660	31392.750
Output per worker (industrial)	397	8267.993	4817.564	689.409	26747.059
Output per worker (services)	397	11814.617	7848.062	689.973	54193.492
total labor force	397	391780.353	474661.500	9500.000	4003800.000
agricultural labor force	397	159541.743	143754.058	1289.262	866953.063
industrial labor force	397	98079.024	190375.450	619.098	1773873.250
service labor force	397	134286.033	217975.016	587.871	2003723.250
Labor force share (aggregate)	397	1.000	0.000	1.000	1.000
Labor force share (agriculture)	397	0.499	0.233	0.013	0.903
Labor force share (industrial)	397	0.209	0.142	0.019	0.616
Labor force share (services)	397	0.294	0.125	0.013	0.795
Earliest decade	397	1822.746	27.630	1800.000	1900.000
Average age of workers	397	26.939	4.070	18.965	36.360
Schooling years per worker (aggregate)	397	2.895	1.929	0.006	6.688
Schooling years per worker (agriculture)	397	2.642	1.749	0.005	6.121
Schooling per worker (industrial)	397	2.930	1.904	0.006	6.597
Schooling per worker (services)	397	3.175	2.056	0.007	7.318
Experience years per worker (aggregate)	316	18.353	2.390	12.673	25.829
Experience years per worker (agriculture)	397	18.298	2.692	12.684	26.130
Experience years per worker (industrial)	397	18.010	2.642	12.643	25.752
Experience years per worker (services)	397	17.764	2.519	12.608	25.248
Human capital per worker (aggregate)	316	2.798	0.587	1.724	4.086
Human capital per worker (agriculture)	397	2.599	0.570	1.722	3.944
Human capital per worker (industrial)	397	2.663	0.611	1.722	4.054
Human capital per worker (services)	397	2.722	0.658	1.722	4.295
Total factor productivity (aggregate)	316	3.820	2.491	1.386	25.286
Total factor productivity (agriculture)	397	119.920	68.189	22.158	590.512
Total factor productivity (industrial)	397	237.353	92.784	44.491	629.997
Total factor productivity (services)	397	182.832	74.798	34.190	677.039
Capital-output ratio (aggregate)	317	500549.808	616979.909	10015.795	5448931.000
Capital-output ratio (agriculture)	397	3.129	3.145	0.349	20.335
Capital-output ratio (industrial)	397	0.810	0.430	0.067	2.688
Capital-output ratio (services)	397	3.041	1.013	0.373	4.810
Physical capital per worker (aggregate)	317	5.383e+09	1.021e+10	72749120.000	1.052e+11
Physical capital per worker (agriculture)	397	12180.067	13404.660	750.147	113077.883
Physical capital per worker (industrial)	397	6960.909	5361.407	228.257	26753.859
Physical capital per worker (services)	394	39445.356	31603.772	1415.218	174613.078

Table 7: Summary statistics (1910-2020)

	N	Mean	SD	Min	Max
Output per worker (aggregate)	591	42667.688	26864.183	4258.951	156677.766
Output per worker (agriculture)	591	37755.701	67186.166	734.797	1065319.000
Output per worker (industrial)	591	41210.265	33544.272	2897.266	336104.313
Output per worker (services)	591	45589.937	24601.807	5438.518	168051.688
total labor force	591	1814456.805	2323630.000	37500.000	18713000.000
agricultural labor force	591	112345.341	134451.071	267.542	866953.063
industrial labor force	591	460759.515	542980.932	6341.799	3608722.250
service labor force	591	1241351.953	1839973.513	21817.900	15684194.000
Labor force share (aggregate)	591	1.000	0.000	1.000	1.000
Labor force share (agriculture)	591	0.128	0.152	0.000	0.713
Labor force share (industrial)	591	0.257	0.101	0.046	0.633
Labor force share (services)	591	0.615	0.163	0.173	0.911
Earliest decade	591	1840.203	38.716	1800.000	1950.000
Average age of workers	591	35.431	3.266	25.833	44.900
Schooling years per worker (aggregate)	591	10.310	2.954	3.069	15.015
Schooling years per worker (agriculture)	591	9.182	2.795	2.589	13.983
Schooling per worker (industrial)	591	10.021	2.758	3.389	15.290
Schooling per worker (services)	591	10.669	2.802	3.620	15.043
Experience years per worker (aggregate)	491	19.223	1.617	12.107	24.038
Experience years per worker (agriculture)	591	20.249	1.786	10.286	25.716
Experience years per worker (industrial)	591	19.410	1.790	10.354	25.323
Experience years per worker (services)	591	18.762	1.759	9.563	24.604
Human capital per worker (aggregate)	491	5.398	1.409	2.579	8.428
Human capital per worker (agriculture)	591	5.308	1.423	2.488	8.738
Human capital per worker (industrial)	591	5.663	1.487	2.637	9.169
Human capital per worker (services)	591	5.958	1.570	2.686	9.295
Total factor productivity (aggregate)	491	3.096	1.232	1.430	11.288
Total factor productivity (agriculture)	591	204.894	196.594	15.823	2488.249
Total factor productivity (industrial)	591	284.220	103.514	43.220	932.710
Total factor productivity (services)	591	250.602	66.970	95.944	622.141
Capital-output ratio (aggregate)	491	2954669.888	4333582.519	63957.289	36522612.000
Capital-output ratio (agriculture)	591	4.337	2.632	0.268	23.091
Capital-output ratio (industrial)	591	1.990	0.809	0.611	7.152
Capital-output ratio (services)	591	3.622	0.581	2.322	5.608
Physical capital per worker (aggregate)	491	1.398e+11	2.830e+11	9.785e+08	2.711e+12
Physical capital per worker (agriculture)	591	146658.962	220110.510	2823.276	1921466.500
Physical capital per worker (industrial)	591	89634.187	116146.844	5643.142	1447031.250
Physical capital per worker (services)	591	158259.550	78155.512	18658.779	620806.375

Table 8: States by year of coverage

State	First decade	Included in 1800	Included in 1850	Included in 1880
AK	1950			
AL	1820		✓	✓
AR	1840		✓	✓
AZ	1880			✓
CA	1850		✓	✓
CO	1880			✓
CT	1800	✓	✓	✓
DE	1800	✓	✓	✓
FL	1840		✓	✓
GA	1800	✓	✓	✓
HI	1950			
IA	1850		✓	✓
ID	1880			✓
IL	1820		✓	✓
IN	1820		✓	✓
KS	1870			✓
KY	1800	✓	✓	✓
LA	1820		✓	✓
MA	1800	✓	✓	✓
MD	1800	✓	✓	✓
ME	1820		✓	✓
MI	1840		✓	✓
MN	1860			✓
MO	1830		✓	✓
MS	1820		✓	✓
MT	1870			✓
NC	1800	✓	✓	✓
ND	1900			
NE	1870			✓
NH	1800	✓	✓	✓
NJ	1800	✓	✓	✓
NM	1850		✓	✓
NV	1870			✓
NY	1800	✓	✓	✓
OH	1810		✓	✓
OK	1920			
OR	1860			✓
PA	1800	✓	✓	✓
RI	1800	✓	✓	✓
SC	1800	✓	✓	✓
SD	1900			
TN	1800	✓	✓	✓
TX	1850		✓	✓
UT	1860			✓
VA	1800	✓	✓	✓
VT	1800	✓	✓	✓
WA	1870			✓
WI	1850		✓	✓

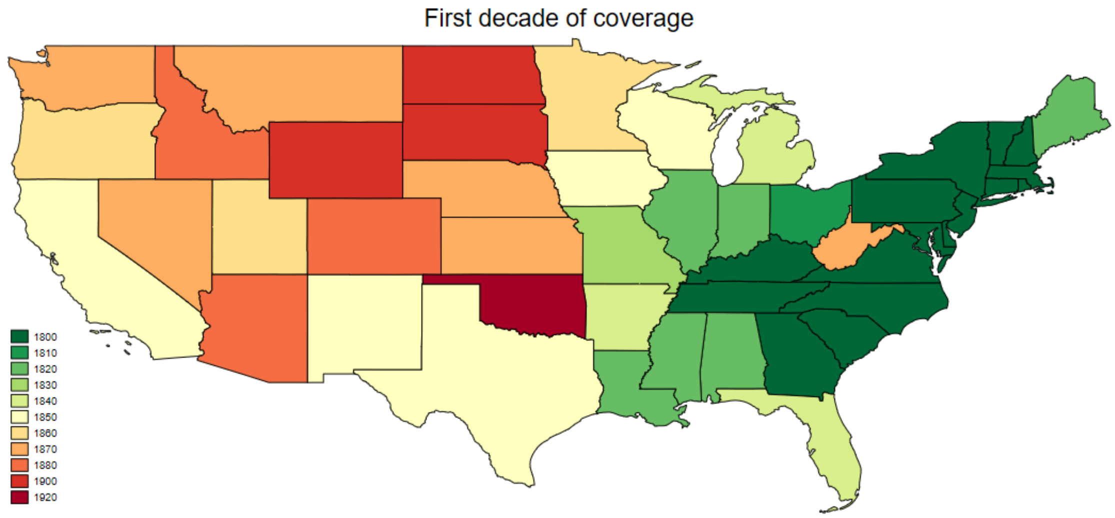

Figure 6: States by year of first observation

A.2 Data Construction

The main dataset comes from Tamura, Witham and Khatri (2025, working). A copy of the original datasets and Stata do files is provided at:

`https://github.com/jbcmpbl/Job-Market-Paper-Replication/tree/main`

A.2.1 Human capital

Human capital was estimated as a standard Mincerian return following the equation:

$$\frac{H_{ijk}}{L_{ijk}} = 0.1S_{ijk} + 0.0495X_{ijk} - 0.0007X_{ijk} \tag{5}$$

in which S_{ijk} corresponds to years of schooling and X_{ijk} to years of education in country i, year j and sector k. Years of schooling was available in the original dataset, while years of experience is calculated as:

$$X_{ijk} = \bar{A}_{ij} - S_{ijk} - 6 \tag{6}$$

for average age A_{ijk}. Average age was not available at the sectoral level and the original dataset only had coverage from 1840 to 2000. For the decades of 1800, 1810, 1820 and 1830, I imputed, at the state-level, each state's 1840 value. For 2010 and 2020 median worker age values were taken from US Census' American Community Survey's 1-year estimates with the missing 2020 year being imputed as the average of 2019 and 2020's values. The original datasets (ACSST1Y2010, ACSST1Y2019, and ACSST1Y2021) can be found at the following links:

`https://data.census.gov/table/ACSST1Y2010.S0101?q=median+age&g=010XX00US$0400000`

`https://data.census.gov/table/ACSST1Y2019.S0101?q=median+age&g=010XX00US$0400000`

`https://data.census.gov/table/ACSST1Y2021.S0101?q=s0101&g=010XX00US$0400000`

A.2.2 Total factor productivity

As a function of human capital, total factor productivity was also recomputed, according to the standard equation:

$$A_{ijk} = \frac{Y_{ijk}}{K_{ijk}^{\alpha} H_{ijk}^{1-\alpha}} \tag{7}$$

in which A_{ijk} represents total factor productivity, K_{ijk} represents physical capital, and H_{ijk} represents human capital in state i, year j and sector k.

A.2.3 Interpolation of capital for 1840

The United States Census of 1840 is widely recognized as problematic; as the first census to systematically calculate stocks of wealth, occupations and factors of production, it contains many inconsistencies. The original dataset from Tamura, Witham and Khatri contains plausible estimations of output per worker, and human capital per worker is calculated anew as described in the previous section. However, for the year 1840 specifically the original dataset contains an implausibly large downturn in physical capital per worker, entirely located within the service sector, that rebounds by 1850, with some states' capital stocks declining by more than ten-fold. Capital per worker in services is not used directly in the presentation of any of this paper's results, but it is noted here that the accompanying dataset contains a log-linear interpolation for services capital per worker for 1840 using the 1830 and 1850 values; for states first covered from 1840 on, capital per worker in services is dropped.

A.2.4 Excluded observations

For the year 1850 California was observed to have an unrealistically high service workforce (79.5%) that, by 1860, collapses to 44.7% and proceeds to grow at a measured and slow pace, only reaching its old level by the 2000s. Given the extreme outlier nature of the observation, its incongruity with the established literature on California's sectoral composition and the use of workforce size to calculate most of the relevant statistics (including output per worker), I dropped this observation entirely as a likely transcription error.

B Robustness

B.1 Alternate Periodizations

As the σ-convergence results do not match with a conventional symmetrical two-period split, in this section I present alternative periodizations as robustness.

B.1.1 Rolling windows

Figure 8 summarizes the results of cross-sectional regressions of specification (1) for each decade and sector with a 95% confidence interval shaded in gray. As the sample size grows over the early nineteenth century, the confidence interval accordingly tightens. In most decades and in most sectors the 95% confidence interval of β sits firmly in the negative space, supporting the alternative hypothesis of smaller economies growing faster. However, services are an exception in the earliest decades and small sample sizes allow for the possibility of β-divergence, which is more cosistent with the observed pattern of σ-divergence in agriculture, services and the aggregate economy to circa 1850. Changes in the pace of convergence are chaotic and inconsistent.

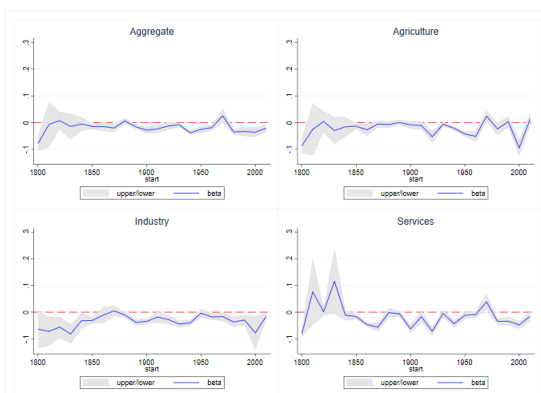

Figure 7: β-convergence coefficients in cross-sectional decadal regressions

In order to expand the sample and smooth the data, Figure 9 presents the results of the same exercise applied to a 40-year rolling window (ie a panel of 1800-1840, a panel of 1810-1850, one of 1980-2020, etc.). When allowing for a rolling window, a much cleaner picture emerges of β-convergence that slows in the early period and quickens afterwards in agriculture and industry and a long reversal from 1900-1990 in services. In this the β-convergence and σ-convergence results differ, as agriculture σ-diverges heavily past 1960 but

β-convergence strengthens, while services σ-converge and β-diverge over the same period.

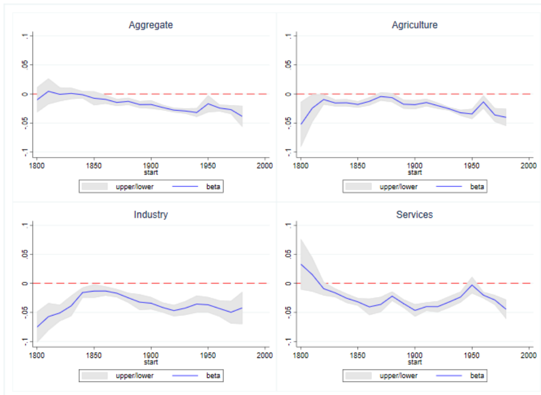

Figure 8: β-convergence coefficients in 40-year rolling window panel regressions

B.1.2 Four periods

Having presented rolling windows, Tables 9-12 report empirical results for specification (1) for a four-period periodization; 1800-1850 (to correspond to the conventional historical periodization of the First Industrial Revolution and American Market Revolution), 1850-1910 (corresponing to the Civil War, Reconstruction and early Progressive Era), 1910-1960 (corresponding to the later Progressive Era), and finally 1960-2020; the choice of decades for each is a compromise between the observed patterns of σ-convergence, historical meaning and balance. When adding the extra granularity of four periods, Table 9's β-coefficient estimates show a pattern more consistent with the σ-convergence results of Figure 5, including no statistical significance for aggregate convergence over 1800-1850 (when σ diverged widely).

	1800-1850	1850-1910	1910-1960	1960-2020
Initial Output per worker (aggregate) (decadal)	-0.0120	-0.0148***	-0.0229***	-0.0208***
	(0.0103)	(0.0031)	(0.0027)	(0.0029)
Observations	107	243	288	200
R^2	0.64	0.23	0.41	0.47

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: β-convergence, 1800-1850, 1850-1910, 1910-1960 and 1960-2020

Table 10, reporting for agriculture, shows statistically significant β-convergence in all periods, but the magnitude weakens past 1960 and the significance itself is weaker in both the beginning and end periods, again tracking the specific patterns observed in agricultural output per worker's standard deviation.

	1800-1850	1850-1910	1910-1960	1960-2020
Initial Output per worker (agriculture) (decadal)	-0.0205**	-0.0088***	-0.0232***	-0.0125
	(0.0090)	(0.0024)	(0.0031)	(0.0082)
Observations	107	243	288	200
R^2	0.60	0.34	0.56	0.55

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: β-convergence in agriculture, 1800-1850, 1850-1910, 1910-1960 and 1960-2020

Table 11, reporting for industry, is the least consistent with the σ-convergence story, showing very strong convergence in the earliest period despite strong σ-divergence. The result is less contradictory when considering the balanced panel to account for the effects of adding new and, generally, heavily agricultural or (due to gold rushes) abnormally productive industrial (mining is counted as industry in Tamura, Witham and Khatri (2025)'s dataset) states. This may be evidence of the importance of leapfrogging (see Section 5) as churn in relative state rank was highest in manufacturing and remained elevated past other sectors for longer.

	1800-1850	1850-1910	1910-1960	1960-2020
Initial Output per worker (industrial) (decadal)	-0.0588***	-0.0215***	-0.0310***	-0.0245***
	(0.0074)	(0.0038)	(0.0050)	(0.0044)
Observations	107	243	288	200
R^2	0.58	0.56	0.67	0.53

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11: β-convergence in industry, 1800-1850, 1850-1910, 1910-1960 and 1960-2020

Lastly, Table 12, reporting for services, shows no evidence of β-convergence for the earliest period (1800-1850), very strong convergence afterwards and then weakening in the trend over the twentieth century, all of which is consistent with what is observed in its σ-convergence patterns reported in Figure 5. Whereas with industry there is a high amount of rank change, in services there was no such trend from the beginning; state's relative positions were comparatively stable over the whole 220-year period, supporting the interpretation that the β-convergence, σ-divergence pattern over the nineteenth century is driven by states frequently overtaking each other.

	1800-1850	1850-1910	1910-1960	1960-2020
Initial Output per worker (services) (decadal)	-0.0060	-0.0331***	-0.0481***	-0.0193***
	(0.0109)	(0.0031)	(0.0048)	(0.0027)
Observations	107	243	288	200
R^2	0.08	0.48	0.68	0.35

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12: β-convergence in services, 1800-1850, 1850-1910, 1910-1960 and 1960-2020

B.1.3 One period

Tables 13-16 report empirical results for specification (1) for a full unbalanced panel and balanced panel and specification (2) for a cross-section of 1800-2020. β-convergence in the single panel regressions remains highly statistically significant across all specifications with a pattern of weaker convergence in agriculture and the strongest convergence in industry.

	1800-2020		
	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (aggregate) (decadal)	-0.0166***		-0.0078***
	(0.0023)		(0.0011)
Initial Output per worker (aggregate) (long-run)		-0.0048***	
		(0.0008)	
Observations	791	16	320
R^2	0.48	0.71	0.54

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13: β-convergence, 1800-2020

	1800-2020		
	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (agriculture) (decadal)	-0.0175***		-0.0230***
	(0.0021)		(0.0035)
Initial Output per worker (agriculture) (long-run)		0.0041	
		(0.0079)	
Observations	791	16	320
R^2	0.56	0.02	0.56

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14: β-convergence in agriculture, 1800-2020

	1800-2020		
	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (industrial) (decadal)	-0.0299***		-0.0285***
	(0.0032)		(0.0054)
Initial Output per worker (industrial) (long-run)		-0.0045***	
		(0.0008)	
Observations	791	16	320
R^2	0.58	0.70	0.60

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 15: β-convergence in industry, 1800-2020

	1800-2020		
	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (services) (decadal)	-0.0286***		-0.0162***
	(0.0030)		(0.0022)
Initial Output per worker (services) (long-run)		-0.0045***	
		(0.0008)	
Observations	791	16	320
R^2	0.43	0.72	0.37

Standard errors in parentheses
Notes:
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 16: β-convergence in services, 1800-2020

B.2 Geographical Variation

B.2.1 Jack-knife estimation

In order to test the robustness of my results with respect to geographical variation and outliers, I conducted jack-knife (leave-one-out regression) estimation at the state level. Table 17 reports the jack-knife estimation results for the baseline specification: specification (1), unbalanced panel, periods of 1800-1910 and 1910-2020. The jack-knife β estimates in each case demonstrate a very tight spread around the same estimates as the full panel, confirming that no single outlier state drives the results presented.

In order to test the robustness of my robustness, I also present results for jack-knifing at the census-divisional level in Table 18. As expected, the standard deviation of the jack-knife estimates doubles to triples for each sector and specification, but the stnadard deviation remains very small relative to the observed means, and the means are almost unchanged. As such, the results are not driven by outlier sub-regions (such as, say, the South Atlantic states or the Mountain West) any more than by a single state.

B.2.2 Regional variation

Lastly, I present robustness at the highest meaningful level of geographical division by presenting separated panels for the South, North and West in which the South is the West South Central, East South Central and South Atlantic states, the North is West North Central, East North Central, Middle Atlantic and New England states, and the West is the Pacific and Mountain states. Table 19 reports results for the aggregate output per worker. Notably, there is considerable variation at the regional level in convergence

Table 17: Jack knife estimation results: β-convergence

	1800-1910				1910-2020			
Sector	Mean β	Std. dev.	Min	Max	Mean β	Std. dev.	Min	Max
Aggregate	-0.014	0.000	-0.016	-0.013	-0.023	0.000	-0.024	-0.021
Agriculture	-0.010	0.000	-0.011	-0.009	-0.021	0.000	-0.022	-0.020
Industry	-0.030	0.001	-0.033	-0.029	-0.030	0.001	-0.032	-0.027
Services	-0.028	0.001	-0.030	-0.027	-0.041	0.001	-0.043	-0.039

rates, with the South having notably weaker convergence (one-third the magnitude in the early period) than the West. This difference in magnitude falls for the North and South for the later period, but the West's convergence remains relatively strong.

Table 18: Jack knife estimation results: β-convergence

	1800-1910				1910-2020			
Sector	Mean β	Std. dev.	Min	Max	Mean β	Std. dev.	Min	Max
Aggregate	-0.015	0.002	-0.020	-0.012	-0.022	0.002	-0.025	-0.019
Agriculture	-0.010	0.001	-0.012	-0.008	-0.021	0.001	-0.023	-0.019
Industry	-0.031	0.002	-0.035	-0.029	-0.030	0.004	-0.032	-0.020
Services	-0.029	0.003	-0.036	-0.026	-0.040	0.002	-0.044	-0.037

	1800-1910			1910-2020		
	North	South	West	North	South	West
Initial Output per worker (aggregate) (decadal)	-0.0278***	-0.0124***	-0.0355***	-0.0150**	-0.0178***	-0.0320***
	(0.0049)	(0.0033)	(0.0110)	(0.0059)	(0.0029)	(0.0060)
Observations	167	139	44	210	158	120
R^2	0.58	0.65	0.59	0.53	0.54	0.51

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 19: β-convergence by major region, 1800-1910 & 1910-2020

Table 20 reports the same empirical test for agriculture. In agriculture, it is the South that has closer parity to the West in convergence speed in both early and late periods and the North shows no regional pattern of convergence in the early period, although all approach an estimate between 2% and 3% in the late period. Regional regressions reveal that the full panel masks patterns of weak convergence in the North relative to the rest of the country.

	1800-1910			1910-2020		
	North	South	West	North	South	West
Initial Output per worker (agriculture) (decadal)	-0.0041	-0.0187**	-0.0209***	-0.0192***	-0.0241***	-0.0283***
	(0.0041)	(0.0064)	(0.0050)	(0.0032)	(0.0056)	(0.0056)
Observations	167	139	44	210	158	120
R^2	0.57	0.68	0.39	0.56	0.65	0.65

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 20: β-convergence in agriculture by major region, 1800-1910 & 1910-2020

Table 21 reports the test for industry. In industry, the North and South demonstrate very high - inconsistent with the traditional benchmark - rates of convergence around 4-6% for the early period, falling to more normal levels in the late period while the West's already high converges strengthens. The strong rate of convergence in industry relative to services remains consistent with the main results presented. Notably, the elevated convergence rates (relative to the nation) in all sectors suggests some evidence for convergence clubs. Preliminary investigation of the existence of state convergence clubs did not yield any evidence of clear regional club formation, but this result suggests that the national panel may mask some regional dynamics.

	1800-1910			1910-2020		
	North	South	West	North	South	West
Initial Output per worker (industrial) (decadal)	-0.0551***	-0.0477***	-0.0341***	-0.0175***	-0.0184***	-0.0472***
	(0.0116)	(0.0066)	(0.0077)	(0.0020)	(0.0062)	(0.0123)
Observations	167	139	44	210	158	120
R^2	0.67	0.71	0.69	0.81	0.70	0.64

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 21: β-convergence in industry by major region, 1800-1910 & 1910-2020

Finally, Table 22 reports the test for services. in services, very strong intraregional convergence is observed in the North and West in the early period while Southern convergence remains comparatively weak until the late period. The interpretation from the main result of service convergence rates eclisping industrial convergence rates in the late period remains intact.

	1800-1910			1910-2020		
	North	South	West	North	South	West
Initial Output per worker (services) (decadal)	-0.0406***	-0.0214***	-0.0516***	-0.0396***	-0.0379***	-0.0330***
	(0.0043)	(0.0033)	(0.0112)	(0.0113)	(0.0052)	(0.0059)
Observations	167	139	44	210	158	120
R^2	0.53	0.52	0.49	0.70	0.67	0.64

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 22: β-convergence in services by major region, 1800-1910 & 1910-2020

Collectively, I find that the regional regressions show additional detail, with the North having been disfavored in agriculture and the South in services (consistent with an interpretation of the North urbanizing early and the South urbanizing late), but the regional robustness does not contradict the main finding.

C Tables

	1800-1910			1910-2020		
	Unbal. panel	Bal. panel	Cross-section	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (aggregate) (decadal)	-0.0145***	-0.0057***		-0.0225***	-0.0222***	
	(0.0031)	(0.0019)		(0.0023)	(0.0025)	
Initial Output per worker (aggregate) (long-run)			0.0028			-0.0080***
			(0.0039)			(0.0005)
Observations	350	176	16	488	470	47
R^2	0.47	0.53	0.04	0.43	0.46	0.87

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 1: β-convergence, 1800-1910 and 1910-2020

	1800-1910			1910-2020		
	Unbal. panel	Bal. panel	Cross-section	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (agriculture) (decadal)	-0.0097***	-0.0176***		-0.0212***	-0.0215***	
	(0.0023)	(0.0040)		(0.0026)	(0.0026)	
Initial Output per worker (agriculture) (long-run)			-0.0174**			-0.0077***
			(0.0067)			(0.0015)
Observations	350	176	16	488	470	47
R^2	0.46	0.49	0.33	0.56	0.56	0.38

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2: β-convergence in agriculture, 1800-1910 and 1910-2020

	1800-1910			1910-2020		
	Unbal. panel	Bal. panel	Cross-section	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (industrial) (decadal)	-0.0304***	-0.0310***		-0.0299***	-0.0298***	
	(0.0040)	(0.0071)		(0.0045)	(0.0047)	
Initial Output per worker (industrial) (long-run)			-0.0025			-0.0113***
			(0.0022)			(0.0010)
Observations	350	176	16	488	470	47
R^2	0.55	0.57	0.08	0.65	0.66	0.74

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: β-convergence in industry, 1800-1910 and 1910-2020

	1800-1910			1910-2020		
	Unbal. panel	Bal. panel	Cross-section	Unbal. panel	Bal. panel	Cross-section
Initial Output per worker (services) (decadal)	-0.0283***	-0.0153***		-0.0406***	-0.0420***	
	(0.0034)	(0.0026)		(0.0040)	(0.0041)	
Initial Output per worker (services) (long-run)			-0.0024			-0.0087***
			(0.0032)			(0.0006)
Observations	350	176	16	488	470	47
R^2	0.32	0.26	0.04	0.61	0.64	0.81

Standard errors in parentheses

Notes:

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: β-convergence in services, 1800-1910 and 1910-2020

D Figures

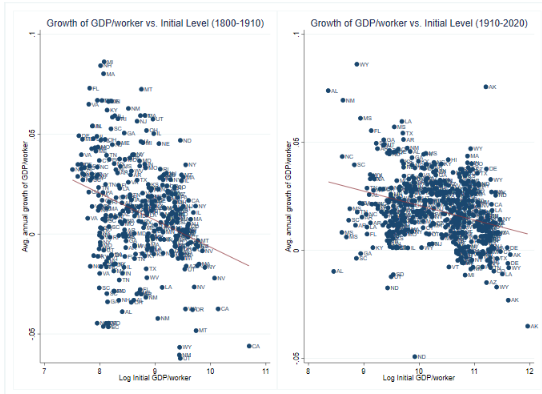

Figure 1: β-convergence scatterplot, decadal intervals, 1800-1910 and 1910-2020

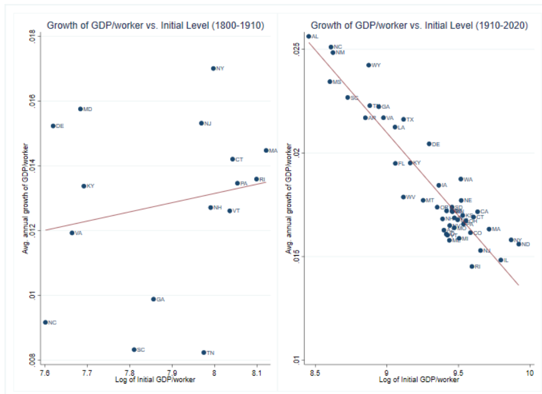

Figure 2: β-convergence scatterplot, long-run, 1800-1910 and 1910-2020

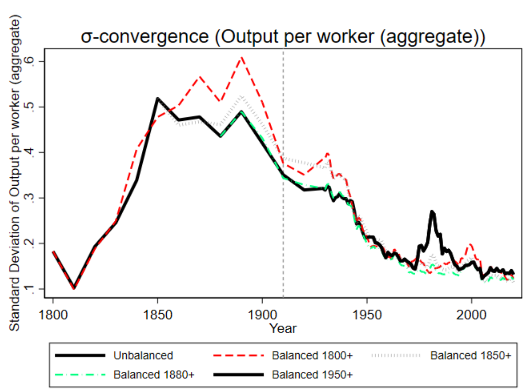

Figure 3: σ-convergence of output per worker for each sector

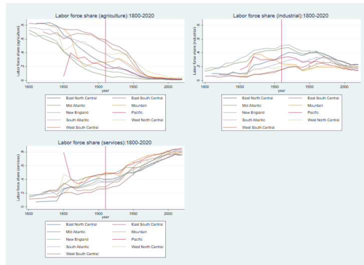

Figure 4: Labor force share in each sector, 1800-2020, Census division level

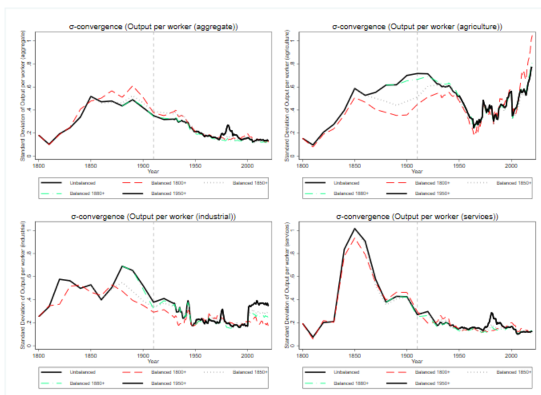

Figure 5: σ-convergence of output per worker for each sector (unbalanced panel)